🏦 Bank Transaction Analyzer

A beginner SQL project analyzing personal bank transaction data to uncover spending patterns, monthly trends, and top expense categories.

> **Tools used:** SQL · SQLite · DB Browser for SQLite
> **Dataset:** Bank Transaction Dataset — Kaggle
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📌 Project Overview

This project explores a bank transactions dataset using SQL queries to answer real business questions:

- Where is most money being spent?
- Which months have the highest spending?
- Who are the top merchants by transaction volume?
- What is the average transaction size by category?

📁 Repository Structure

```
bank-transaction-analyzer/
│
├── data/
│   └── transactions.csv          # Source dataset
│
├── queries/
│   └── analysis_queries.sql      # All SQL queries used in analysis
```

```
│
├── images/
│   └── spending_by_category.png  # Screenshot of key results
│
└── README.md
```

🔍 Key Findings

> *(Fill these in after running your queries — 3 to 5 bullet points)*

- 💳 **Food & Dining** accounted for the largest share of spending at **X%**
- 📅 **Month X** had the highest total transactions at **¥X,XXX**
- 🏪 Top merchant by transaction count was **[Merchant Name]**
- 💴 Average transaction size was **¥X,XXX** across all categories

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🗂️ SQL Queries

1. Total Spending by Category

```sql
SELECT category,
       COUNT(*) AS transaction_count,
       ROUND(SUM(amount), 2) AS total_spent
FROM transactions
GROUP BY category
ORDER BY total_spent DESC;
```

2. Monthly Spending Trend

```sql
SELECT strftime('%Y-%m', date) AS month,
       ROUND(SUM(amount), 2) AS monthly_total
FROM transactions
GROUP BY month
ORDER BY month;
```

3. Top 5 Merchants by Volume

```sql
SELECT merchant,
       COUNT(*) AS visits,
       ROUND(SUM(amount), 2) AS total_spent
FROM transactions
GROUP BY merchant
ORDER BY total_spent DESC
LIMIT 5;
```

4. Average Transaction Size by Category

```sql
SELECT category,
       ROUND(AVG(amount), 2) AS avg_transaction
FROM transactions
GROUP BY category
ORDER BY avg_transaction DESC;
```

5. Largest Single Transactions

```sql
SELECT date, merchant, category, amount
FROM transactions
ORDER BY amount DESC
LIMIT 10;
```

🛠 How to Run This Project

1. Download the dataset from the Kaggle link above
2. Install DB Browser for SQLite (free)
3. Open DB Browser → Import the CSV as a new table named `transactions`
4. Open `queries/analysis_queries.sql` and run each query
5. Screenshot your results and save to the `/images` folder

📚 What I Learned

- Writing SQL queries using `SELECT`, `GROUP BY`, `ORDER BY`, `LIMIT`
- Using aggregate functions: `SUM()`, `COUNT()`, `AVG()`, `ROUND()`
- Using `strftime()` to extract month from date fields in SQLite
- Structuring a data project for GitHub

🔮 Next Steps

- [ ] Add Python visualizations using Pandas + Matplotlib
- [ ] Connect SQLite to Jupyter Notebook via `pd.read_sql()`
- [ ] Expand analysis with a fraud detection angle

📫 Connect

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